November Examinations Mathematics

WHAT YOU NEED TO KNOW

BOTH PAPERS – 2 hours (100-111 marks)

Divided into Section A and Section B

MATHEMATICS PAPER 1

Algebraic Expressions HINT: SIMPLIFY/FACTORISE

- The number System
- Notation and representing numbers
- Surds
- Simplification of Algebraic Expressions
- Factorisation
- Fractions

Equations and Inequalities HINT: SOLVE

- Linear Equations
- Fractions in equations (restrictions)
- Literal equations
- Quadratic Equations
- Simultaneous equations
- Exponential Equations
- Linear Inequalities (linked to notation)
- Applications

Exponents

- Laws of exponents
- Simplification of expressions using rules, by applying prime factors, by factorising
- Exponential equations

Number Patterns

- linear patterns
- other by inspection
- finding the rule Tn and finding n
- applications

Functions – see summary material handed out this week

Straight lines (all applicable to analytical geometry as well)

Quadratic Function

Hyperbola

Exponential graph

- Functional notation
- Sketch all graphs
- Find equations of all graphs
- Graph interpretation (types of questions applicable to trig graphs in p2 as well)

Financial Mathematics

- Simple interest (hire purchase)
- Compound interest with annual interest rates only (and inflation)
- Conversions
- Complex timeline questions

Probability

- Probability scale
- Calculating probability
- Venn Diagrams
- Use of $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
- Mutually exclusive events, independent events, Complementary events, Exhaustive events
- Tree diagrams with and without replacement